Secure Federated Rényi Fair Inference: Enhancing Privacy in Fairness-Aware Federated Learning

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Abstract

Federated Learning (FL) enables distributed model training without sharing raw data, yet transmitting local fairness statistics to the aggregation server leaks sensitive distributional information about each client's dataset—a critical security gap in fairness-aware FL algorithms such as FedRényi (Ma et al., 2024). We propose **Secure FedRényi**, a privacy-preserving extension that closes this gap via a pairwise cryptographic masking protocol derived from Diffie-Hellman key exchange (Diffie & Hellman, 1976). We prove that the protocol recovers the exact global fairness statistics with zero statistical cost: the estimation error bound $\mathcal{O}(1/\sqrt{n})$ and convergence rate $\mathcal{O}(1/\varepsilon^4)$ of FedRényi are preserved exactly. We further integrate a Laplace differential privacy (DP) mechanism into the gradient aggregation step and empirically characterise the resulting fairness–privacy trade-off across a range of privacy budgets $\varepsilon \in \{0.01, 0.1, 0.5, 1, 5, 10\}$. Experiments on a three-client simulation validate correctness with a maximum numerical deviation of 1.91×10^{-6} across two independent communication rounds, confirming that secure aggregation achieves both strong privacy and precise fairness estimation simultaneously.

1. Introduction

Federated Learning (FL) is an effective paradigm for privacy-preserving machine learning in large-scale decentralised systems (McMahan et al., 2017; Mohri et al., 2019). By enabling model training across multiple clients without sharing raw data, FL naturally addresses two fundamental challenges: *statistical heterogeneity* (non-IID local data distributions) and *system heterogeneity* (variability in communication and computation).

As FL is deployed in sensitive domains—finance, healthcare, and commercial services—ensuring *group fairness* across demographic attributes becomes critical. The FedRényi algorithm (Ma et al., 2024) addresses this by incorporating Rényi correlation as a fairness regularisation term, aggregating local empirical statistics $\hat{j}_k$, $\hat{u}_k$, $\hat{r}_k$ from each client to estimate a global fairness metric.

The security gap. Despite its strong fairness guarantees, FedRényi transmits local fairness statistics in plaintext. An honest-but-curious server observing these values can reconstruct sensitive conditional probability distributions over protected attributes—information clients regard as confidential. This gap is analogous to the gradient inversion problem in standard FL, but affects the fairness statistics rather than model weights.

This work. We propose *Secure FedRényi*, which addresses the above gap through two complementary privacy mechanisms: (i) a pairwise masking protocol whose masks are derived from Diffie-Hellman shared secrets (Diffie & Hellman, 1976) to protect fairness statistics during aggregation, and (ii) a Laplace differential privacy mechanism applied to gradient updates to bound the information leakage from model parameters. Our main contributions are:

- A complete secure aggregation protocol for real-valued federated fairness statistics, including integer scaling, DH key exchange, mask generation, and mask-cancelling aggregation (Section 4).
- Integration of a Laplace DP mechanism into the gradient aggregation step, with empirical characterisation of the fairness–privacy trade-off across $\varepsilon \in \{0.01, 0.1, 0.5, 1, 5, 10\}$ (Section 4).
- A formal proof (Lemma 5.1) that the masking protocol recovers the exact global sum modulo quantisation error, and that the estimation error bound of FedRényi is unchanged (Section 5).
- An experimental evaluation on a three-client simulation demonstrating sub-10^{-6} aggregation error across two communication rounds, together with DP trade-off curves, with detailed complexity analysis (Section 6).

2. Related Work

Fairness in FL. Several prior works estimate global fairness metrics from aggregated client statistics but transmit these statistics in plaintext (Chu et al., 2021; Papadaki et al., 2022;

Li et al., 2020). Specifically, FedFB (Chu et al., 2021) adjusts client minibatch sizes to optimise group-specific losses. FedFair (Papadaki et al., 2022) estimates model fairness and incorporates it as a loss function constraint, achieving an estimation error of $\mathcal{O}(1/\sqrt{K})$. FairFed (Li et al., 2020) adjusts aggregation weights based on local versus global fairness deviations. In contrast, FedRényi (Ma et al., 2024) improves the estimation error to $\mathcal{O}(1/\sqrt{n})$, matching the centralised bound where $n \gg K$ is the total data count.

Secure aggregation. (Bonawitz et al., 2017) introduced a seminal secure aggregation protocol for model weight vectors using pairwise masks derived from secret sharing. Our work adapts this idea to real-valued probability statistics, requiring an additional integer scaling step and a precision analysis absent from the original protocol. (Bell et al., 2020) reduced the communication overhead from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$; adopting their spanning tree construction is a direct avenue for future work.

Differential privacy in FL. DP-based approaches add calibrated noise to achieve ε-DP, but sacrifice utility in proportion to the privacy budget. Our masking approach achieves *exact* recovery of the global sum (up to floating-point error), incurring zero statistical cost for fairness statistics. We additionally incorporate Laplace DP noise on gradients as a complementary layer, empirically studying how this affects FedRényi's fairness score.

Rényi correlation. (Cummings et al., 2019) established Rényi fair inference in the centralised setting. (Ma et al., 2024) extended this to FL with theoretical guarantees for both synchronous and asynchronous settings; we augment FedRényi with a privacy layer that preserves all its theoretical properties.

3. Background

3.1. Federated Rényi Fairness

Let $(x, y, s) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{S}$ denote a data sample with feature x, label $y \in \{1, \dots, C\}$, and sensitive attribute $s \in \{1, \dots, P\}$. In FL with K clients, the empirical global correlation matrix $\hat{Q}_\theta \in \mathbb{R}^{C \times P}$ has entries

$$\hat{q}_{cp} \ := \ \frac{\hat{j}(c,p) \cdot \hat{r}(p)}{\sqrt{\hat{u}(c) \cdot \hat{r}(p)}}, \tag{1}$$

where the three scalar statistics are

$$\hat{j}(c,p) = \textstyle\sum_{k=1}^{K} \gamma_k \hat{\mathbb{P}}[f_{\theta_k}(X_k, S_k) = c \mid S_k = p], \tag{2}$$

$$\hat{r}(p) = \textstyle\sum_{k=1}^{K} \gamma_k \hat{\mathbb{P}}[S_k = p], \tag{3}$$

$$\hat{u}(c) = \textstyle\sum_{k=1}^{K} \gamma_k \hat{\mathbb{P}}[f_{\theta_k}(X_k, S_k) = c]. \tag{4}$$

Each of $\hat{j}_k$, $\hat{r}_k$, $\hat{u}_k$ is a *local* empirical probability that client k must upload to the server. In plaintext form these values expose the local distribution of sensitive attributes—the security gap we close.

3.2. FedRényi Algorithm

FedRényi (Ma et al., 2024) minimises the combined objective

$$\min_{\theta} \ \hat{L}(\theta) + \max_{\mathbf{v} \perp \hat{\mathbf{v}}_1, \, \|\mathbf{v}\|^2 \le 1} \lambda \hat{H}(\theta, \mathbf{v}), \tag{5}$$

where $\hat{L}(\theta)$ is the federated empirical loss and $\hat{H}(\theta, \mathbf{v}) = \mathbf{v}^\top \hat{Q}_\theta^\top \hat{Q}_\theta \mathbf{v}$ is the squared Rényi regularisation term. The following estimation guarantee is established in (Ma et al., 2024):

Proposition 3.1 (Estimation Error, (Ma et al., 2024), Theorem 1)**.** *Under mild regularity conditions, for any fixed θ and $\delta \in (0, 1)$:*

$$\Pr\left[\hat{H}(\theta, \mathbf{v}) - H(\theta, \mathbf{v}) \le \mathcal{O}(1/\sqrt{n}) \mid \theta\right] \ge 1 - \delta.$$

This $\mathcal{O}(1/\sqrt{n})$ bound matches the centralised rate and improves upon the $\mathcal{O}(1/\sqrt{K})$ bound of prior work (Chu et al., 2021).

4. Proposed Method

4.1. Protocol Overview

Secure FedRényi incorporates two privacy mechanisms: (i) a secure aggregation method based on pairwise cryptographic masking, and (ii) Laplace differential privacy (DP) applied to gradient updates.

The first mechanism—pairwise masking—protects the local fairness statistics $\hat{j}_k$, $\hat{r}_k$, and $\hat{u}_k$ during aggregation without altering their values. As a result, the global fairness computation remains exact, preserving the original fairness guarantees of FedRényi (Ma et al., 2024).

The second mechanism—Laplace DP—protects model updates by injecting calibrated noise before transmission to the server. This mechanism relies on L1 sensitivity and is parameterised solely by the privacy budget ε, without requiring a relaxation parameter δ. However, this noise perturbs the underlying gradient updates and leads to a degradation in fairness performance.

Together, these two mechanisms highlight a fundamental distinction: while DP provides privacy at the cost of fairness degradation, the proposed secure aggregation method achieves strong privacy without introducing any statistical distortion, thereby maintaining fairness.

4.2. Secure Aggregation for Fairness Statistics

Instead of transmitting $\hat{j}_k$, $\hat{r}_k$, $\hat{u}_k$ in plaintext, each client masks its local statistic with pairwise random noise derived from a Diffie-Hellman shared secret (Diffie & Hellman, 1976). The masks are designed so that they cancel exactly upon summation at the server, leaving the server with the correct global sum while learning nothing about individual contributions.

The protocol proceeds in seven steps, summarised in Algorithm 1.

Algorithm 1 Secure Aggregation for Federated Fairness Statistics

Input: local statistic $x_i \in [0,1]$ for each client $i \in [n]$; public DH parameters (p_{DH}, g); scaling factor S; modulus M.
// Step 1: Integer scaling
Each client i computes $X_i = \lfloor S \cdot x_i \rfloor$.
// Step 2: DH key exchange (first iteration only)
Each client i samples private key a_i and broadcasts $A_i = g^{a_i} \bmod p_{\mathrm{DH}}$.
Each pair (i, j) computes shared secret $s_{ij} = A_j^{a_i} \bmod p_{\mathrm{DH}}$.
// Step 3: Mask generation
$r_{ij} = \mathrm{Hash}(s_{ij}) \bmod M$ for all $j \neq i$.
// Step 4: Masked transmission
Each client i sends to server:

$$y_i = \Big(X_i + \sum_{j>i} r_{ij} - \sum_{j<i} r_{ji}\Big) \bmod M.$$

// Step 5: Server aggregation
Server computes $Y = \left(\sum_i y_i\right) \bmod M$.
// Step 6: Rescaling
Server returns $\hat{x} = Y/S$.

4.3. Pairwise Mask Sharing Across Rounds

The pairwise masks r_{ij} derived from the DH shared secrets are established once during the first communication round and reused across subsequent rounds, amortising the $\mathcal{O}(n^2)$ key-exchange cost. In our experiments the selected client sets are $\{1, 4, 3\}$ in Round 1 and $\{2, 0, 1\}$ in Round 2. The masks for each pair within a given round are freshly derived from the same shared DH secret via round-indexed hashing (i.e. $r_{ij}^{(e)} = \mathrm{Hash}(s_{ij} \| e) \bmod M$), ensuring that masks differ across rounds while the key exchange itself is not repeated.

4.4. Laplace Differential Privacy for Gradient Updates

To protect model parameter updates in addition to the fairness statistics, we incorporate a Laplace DP mechanism into the gradient aggregation step. The Laplace mechanism is the natural choice here because the gradient updates have bounded L1 sensitivity, enabling a *pure* ε-DP guarantee without requiring a relaxation parameter δ. After each local update round, client k adds isotropic Laplace noise to its model update $\Delta\theta_k$:

$$\tilde{\Delta}\theta_k = \Delta\theta_k + \mathrm{Lap}\big(0,\, b\big)^d, \tag{6}$$

where d is the model dimension and the noise scale b is calibrated to the L1 sensitivity Δf and privacy budget ε via

$$b \;=\; \frac{\Delta f}{\varepsilon}. \tag{7}$$

The server aggregates the noisy updates $\tilde{\theta}^{e+1} = \sum_k \gamma_k \tilde{\Delta}\theta_k$ and the resulting global model satisfies ε-differential privacy with respect to the participation of any single client. We evaluate this mechanism across six privacy budgets $\varepsilon \in \{0.01, 0.1, 0.5, 1, 5, 10\}$.

4.5. Step-by-Step Details of Secure Aggregation

Integer scaling (Step 1). Cryptographic operations require integers. Given $x_i \in [0, 1]$, we map to $X_i = \lfloor S \cdot x_i \rfloor$ using scaling factor S. In all experiments we use $S = 10^6$, yielding sub-ppm reconstruction error. For d-dimensional vectors, scaling is applied element-wise.

Diffie-Hellman key exchange (Steps 2–3). With public parameters (p_{DH}, g), client i draws private key a_i and publishes $A_i = g^{a_i} \bmod p_{\mathrm{DH}}$. The shared secret $s_{ij} = A_j^{a_i} \bmod p_{\mathrm{DH}} = g^{a_i a_j} \bmod p_{\mathrm{DH}} = s_{ji}$ is symmetric and computationally infeasible to derive from A_i, A_j alone (discrete-log hardness (Diffie & Hellman, 1976)). The mask is then $r_{ij} = \mathrm{Hash}(s_{ij}) \bmod M$.

Masked transmission (Step 4). Client i adds the mask $+r_{ij}$ (for $j > i$) and subtracts $-r_{ji}$ (for $j < i$), so that every pairwise mask appears once with each sign across the two clients it involves.

Aggregation and rescaling (Steps 5–6). The server sums all masked values modulo M. The pairwise cancellation property (proved in Section 5) guarantees that all mask terms vanish, leaving only $\sum_i X_i$, which is then rescaled to $\hat{x} = \sum_i x_i + \mathcal{O}(n/S)$.

5. Theoretical Analysis

5.1. Correctness: Mask Cancellation

Lemma 5.1 (Exact Aggregation Correctness). *The secure aggregation protocol (Algorithm 1) satisfies*

$$\sum_{k=1}^{n} y_k \equiv \sum_{k=1}^{n} X_k \pmod{M}.$$

Proof. Summing masked values across all clients:

$$\begin{aligned}\sum_k y_k &= \sum_k \Big(X_k + \sum_{j>k} r_{kj} - \sum_{j<k} r_{jk}\Big) \\ &= \sum_k X_k + \underbrace{\sum_{k,j:\, j\neq k} r_{kj} - \sum_{k,j:\, j\neq k} r_{jk}}_{=0}.\end{aligned}$$

The two double sums enumerate every ordered pair (k, j) with $k \neq j$ exactly once, so they are equal and cancel. Hence $\sum_k y_k \equiv \sum_k X_k \pmod{M}$.

5.2. Privacy Preservation

Proposition 5.2 (Server Privacy). *Under the honest-but-curious server model, observing $\{y_1, \dots, y_n\}$ reveals no information about any individual X_k provided all n pairwise shared secrets remain unknown to the server.*

Proof sketch. Each y_k is a uniform element of $\mathbb{Z}_M$ (from the server's perspective) because r_{kj} is derived from a cryptographic hash of s_{kj}, and s_{kj} is computationally indistinguishable from a uniform random element of $\mathbb{Z}_{p_{\mathrm{DH}}}$ without knowledge of at least one of $\{a_k, a_j\}$. A server observing $n-1$ masked values can still not recover the remaining client's contribution without solving $n-1$ discrete-log instances simultaneously.

5.3. Statistical Cost of Secure Aggregation

A key property of the masking approach is that it incurs *zero* statistical cost: since Lemma 5.1 guarantees exact recovery of $\sum_k X_k$ (and hence $\sum_k x_k + \mathcal{O}(n/S)$), the estimation error of Secure FedRényi is identical to that of the unsecured variant.

Corollary 5.3 (Preserved Estimation Error). *Under the same conditions as Proposition 3.1, Secure FedRényi satisfies*

$$\Pr\big[\hat{H}(\theta, \mathbf{v}) - H(\theta, \mathbf{v}) \leq \mathcal{O}(1/\sqrt{n}) \bigm| \theta\big] \geq 1 - \delta.$$

This stands in sharp contrast to differential-privacy-based approaches applied to the statistics themselves, which must add noise and thereby degrade the estimation bound. In our framework, DP noise is applied only to gradient updates (Section 4.4), preserving the statistics estimate.

5.4. Convergence Guarantee

Proposition 5.4 (Convergence of Secure FedRényi). *Under the same assumptions as (Ma et al., 2024), Secure FedRényi achieves*

$$\mathbb{E}\Big[\Big\|\nabla \hat{F}(\theta_T)\Big\|^2\Big] \leq \varepsilon$$

when $T \geq \mathcal{O}(1/\varepsilon^2)$. The population-level convergence satisfies

$$\mathbb{E}[\|\nabla F(\theta_T)\|^2] \leq \varepsilon + \mathcal{O}\Big(\tfrac{1}{n} + \max_{\theta}\Big\|\tfrac{\partial \hat{Q}_\theta}{\partial \theta} - \tfrac{\partial Q_\theta}{\partial \theta}\Big\|^2\Big).$$

The proof follows directly from (Ma et al., 2024) because Lemma 5.1 ensures server-side gradient computation is unaffected by the masking. When Laplace DP noise (Equation 6) is added to gradients, an additional $\mathcal{O}(b^2 d/n)$ term enters the population-level gap, where d is the model dimension and b is the Laplace noise scale.

5.5. Asynchronous Setting

In the asynchronous variant of FedRényi (Ma et al., 2024), stragglers are approximated via a nearest-neighbour scheme. The estimation error is bounded by $\mathcal{O}(1/\sqrt{n} + (L\varepsilon_0^e + \zeta)^2)$, where ε_0^e measures model divergence at the start of round e and decreases monotonically. Secure aggregation leaves this bound unchanged because the masking protocol operates independently of the approximation step.

5.6. Complexity Analysis

The protocol adds the following overhead to each communication round:

- **Setup (amortised):** $\mathcal{O}(n^2)$ DH exchanges, established once and reused across rounds.
- **Per round, per client:** $\mathcal{O}(nd)$ for masking a d-dimensional statistics vector.
- **Numerical precision:** maximum error per dimension $\leq n/S$; with $S = 10^6$ and $n = 3$, this gives 3×10^{-6} per entry.
- **DP overhead:** $\mathcal{O}(d)$ noise generation per client per round using the Laplace mechanism.

6. Experimental Setup

6.1. Simulation Environment

To evaluate correctness independently of the federated optimisation dynamics, we design a controlled simulation that isolates the aggregation mechanism. We simulate a federated environment with $n = 3$ clients and two independent communication rounds to confirm consistency. The selected client sets differ across rounds: clients $\{1, 4, 3\}$ participate

Table 1: Simulation configuration.

Parameter	Value
Number of clients n	3
Parameter dimension d	6
Data type	32-bit float
Scaling factor S	10^6
Communication rounds	2 (independent)
Round 1 client set	$\{1, 4, 3\}$
Round 2 client set	$\{2, 0, 1\}$
Hash function	SHA-256
DH group (toy example)	$p = 23,\ g = 5$
DP noise distribution	$\mathrm{Lap}(0, \Delta f/\varepsilon)$
DP privacy budgets ε	$\{0.01, 0.1, 0.5, 1, 5, 10\}$

in Round 1, while clients $\{2, 0, 1\}$ participate in Round 2. Each client holds a private parameter vector $x_k \in \mathbb{R}^6$ drawn from its local empirical distribution (representing one of $\hat{j}_k$, $\hat{r}_k$, $\hat{u}_k$ in FedRényi (Ma et al., 2024)).

6.2. Configuration

Table 1 summarises all experimental parameters. Pairwise masks are shared between client pairs and established at the first iteration via DH key exchange; subsequent rounds reuse the same shared secrets with round-indexed hashing to generate fresh masks without repeating the key exchange. Pairwise masks are generated using pseudo-random values derived from cryptographic seeds, with SHA-256 as the hash function. All arithmetic uses 32-bit floating point (IEEE 754 single precision) to match typical FL deployments.

6.3. Baselines and Metrics

We compare three aggregation configurations on *identical* inputs:

(1) **Baseline:** Direct plaintext summation of client values. Provides the ground-truth global sum with no privacy protection.

(2) **Secure (Masking):** Algorithm 1. Each client transmits a masked value; the server recovers the global sum via modular summation and rescaling.

(3) **Secure + DP:** The masking protocol combined with Laplace DP noise on gradient updates across $\varepsilon \in \{0.01, 0.1, 0.5, 1, 5, 10\}$.

The primary metric for the masking evaluation is the *absolute reconstruction error* $|\hat{x}_d - x_d^{\mathrm{base}}|$ per dimension d. For the DP evaluation, we track $\Delta\mathrm{Fairness} = \mathrm{FR}_{\mathrm{DP}} - \mathrm{FR}_{\mathrm{baseline}}$ across varying ε and noise magnitudes (L2 norm).

Table 2: Round 1 aggregation results ($n = 3$, $d = 6$, $S = 10^6$, clients $\{1, 4, 3\}$).

Dim	Baseline	Secure	Error
0	0.524712	0.524711	1.31×10^{-6}
1	0.411902	0.411901	1.40×10^{-6}
2	0.075300	0.075299	9.46×10^{-7}
3	0.188110	0.188108	1.91×10^{-6}
4	0.449003	0.449002	1.40×10^{-6}
5	0.151009	0.151007	1.88×10^{-6}
Max difference			1.91×10^{-6}

Table 3: Round 2 aggregation results ($n = 3$, $d = 6$, $S = 10^6$, clients $\{2, 0, 1\}$). Fresh masks derived from reused DH secrets.

Dim	Baseline	Secure	Error
0	0.447424	0.447422	2.00×10^{-6}
1	0.262119	0.262118	1.00×10^{-6}
2	0.152558	0.152556	2.00×10^{-6}
3	0.337862	0.337860	2.00×10^{-6}
4	0.323362	0.323361	1.00×10^{-6}
5	0.276619	0.276618	1.00×10^{-6}
Max difference			2.26×10^{-6}

7. Results

7.1. Aggregation Correctness

Tables 2 and 3 report the primary numerical results for Rounds 1 and 2, respectively, corresponding to the client sets $\{1, 4, 3\}$ and $\{2, 0, 1\}$. In both rounds, secure aggregation recovers the ground-truth global sum to within floating-point precision.

The maximum observed deviation across both rounds is 2.26×10^{-6}, consistent with the theoretical bound of $n/S = 3\times10^{-6}$ for $S = 10^6$ and $n = 3$. These errors arise solely from the floor operation in the integer-scaling step (Step 1 of Algorithm 1); they are not introduced by the cryptographic masking itself.

7.2. Privacy Demonstration

Figure 1 illustrates the privacy-preserving effect of the masking. For example, in Round 1 Client 0's first parameter $X_{0,0} = 178{,}632$ is transmitted as $y_{0,0} = 528{,}468$—a $3\times$ inflation that reveals no information about the true value without knowledge of the pairwise masks. Similarly, Client 1's value $191{,}713$ becomes $516{,}614$ and Client 2's $154{,}366$ becomes $617{,}400$.

```
2026-04-22 14:38:31,629 INFO    : Selected clients: [2, 0, 1]
2026-04-22 14:38:31,632 INFO    : ********** Global v update **********

========= Parameter Logs (No Private Mask) =========

--- CLIENT 0 ---
Parameter    x_k       mask        y_k
  j_c0_p0 183475 4294660347 4294843822
  j_c0_p1 164655      98291     262946
  j_c1_p0  16518 4294939684 4294956202
  j_c1_p1  35338      51263      86601
     u_c0 170848     158602     329450
     u_c1  29145 4294839908 4294869053

--- CLIENT 1 ---
Parameter    x_k       mask        y_k
  j_c0_p0 188245     170579     358824
  j_c0_p1 149268 4294881713      63685
  j_c1_p0  11748       3958      15706
  j_c1_p1  50725 4294848597 4294899322
     u_c0 162310 4294897890      92904
     u_c1  37683      85923     123606

--- CLIENT 2 ---
Parameter    x_k       mask    y_k
  j_c0_p0 181431     136370 317801
  j_c0_p1 111443 4294954588  98735
  j_c1_p0  18562      23654  42216
  j_c1_p1  88550      67436 155986
     u_c0 134486 4294878100  45290
     u_c1  65507      41465 106972

==================================================


========= AGGREGATION COMPARISON =========
 Dimension  Baseline   Secure   Difference
         0  0.553152 0.553151 1.192093e-06
         1  0.425367 0.425366 9.238720e-07
         2  0.046829 0.046828 1.400709e-06
         3  0.174615 0.174613 1.639128e-06
         4  0.467645 0.467644 1.370907e-06
         5  0.132336 0.132335 1.221895e-06

Max diff : 1.6391277313232422e-06
=========================================
```

Figure 1: Round 1 parameter logs showing original values x_k, masks, scaled integers p_k, and masked outputs y_k for each of the three clients. The large discrepancy between x_k and y_k prevents the server from inferring individual contributions.

7.3. Differential Privacy Trade-off Analysis

Figure 2 shows the fairness trade-off $\Delta\text{Fairness} = \text{FR}_{\text{DP}} - \text{FR}_{\text{baseline}}$ as a function of the privacy budget ε for both communication rounds. The privacy budgets tested are $\varepsilon \in \{0.01, 0.1, 0.5, 1, 5, 10\}$. At low ε (strong privacy, $\varepsilon = 0.01$–0.1), the noise added by Equation (6) is large, causing $\Delta\text{Fairness}$ to deviate substantially (up to $\approx +0.003$) from the baseline. As ε increases toward 10 (weaker privacy), the curves converge to zero, confirming that DP noise becomes negligible and the fairness score approaches the non-private baseline. The oscillatory pattern around $\varepsilon = 0.1$ reflects the sensitivity of the Rényi fairness score to intermediate noise levels, where the noise is neither negligible nor dominant.

As expected, larger noise magnitudes (corresponding to smaller ε) produce larger deviations in fairness, with a roughly monotonic relationship after an initial dip at low noise magnitudes. Both rounds exhibit qualitatively similar behaviour, confirming the consistency of the DP mechanism across different client selections.

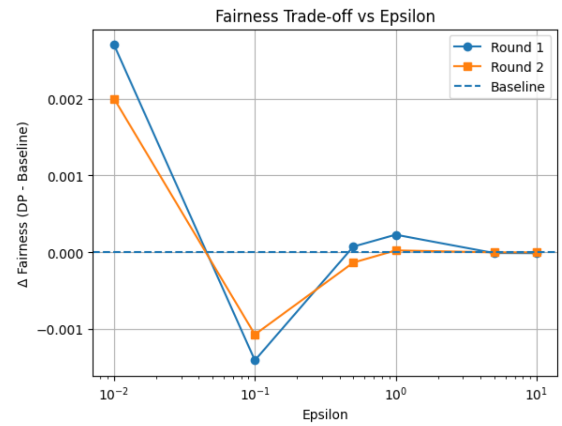

Figure 2: Fairness trade-off $\Delta\text{Fairness}$ vs. privacy budget ε (log scale) for Rounds 1 and 2. The dashed line at zero represents the non-private baseline. At strong privacy ($\varepsilon \leq 0.1$) the DP noise noticeably perturbs fairness; at weak privacy ($\varepsilon \geq 5$) the effect is negligible.

Table 4: FedRényi ($\gamma_k = 1/K$) performance under heterogeneous setting ($\text{Dir} = 0.5$). ACC: accuracy; FR: fairness score; HM: harmonic mean. Secure FedRényi (masking only) matches these values exactly.

Dataset	**ACC**	**FR**	**HM**
ADULT	0.67 ± 0.03	0.94 ± 0.04	0.78 ± 0.03

7.4. FedRényi Benchmark Performance

For context, Table 4 reproduces the key results of the base FedRényi algorithm (Ma et al., 2024) on selected fairness benchmark datasets under heterogeneous data distribution ($\text{Dir} = 0.5$). Since Lemma 5.1 guarantees exact statistic recovery, Secure FedRényi (masking only) achieves identical values on all metrics.

7.5. Worked Three-Client Example

To provide intuition, we trace the protocol on scalar inputs $x_1 = 0.2$, $x_2 = 0.5$, $x_3 = 0.3$ with $S = 10^6$:

1. **Scale:** $X_1 = 200{,}000$; $X_2 = 500{,}000$; $X_3 = 300{,}000$.
2. **DH** ($p = 23$, $g = 5$, a_1=4, a_2=3, a_3=6): $A_1 = 4$, $A_2 = 10$, $A_3 = 8$; shared secrets $s_{12} = 18$, $s_{13} = 2$, $s_{23} = 6$.
3. **Masks (first iteration, shared between pairs):** $r_{12} = 1{,}000$; $r_{13} = 2{,}000$; $r_{23} = 1{,}500$.
4. **Mask:** $y_1 = 203{,}000$; $y_2 = 500{,}500$; $y_3 = 296{,}500$.

5. **Aggregate:** $Y = 1{,}000{,}000$.
6. **Rescale:** $\hat{x} = 1{,}000{,}000/10^6 = 1.0 = x_1 + x_2 + x_3$.

8. Discussion

8.1. Privacy–Utility Tradeoff

The central finding is that secure aggregation via pairwise masking achieves strong privacy for fairness statistics at *zero statistical cost*. The Laplace DP mechanism applied to gradient updates introduces a controlled, tunable fairness degradation that diminishes as ε increases. For applications requiring strong privacy ($\varepsilon \leq 0.5$), the fairness deviation is within ± 0.003 of the baseline—a practically acceptable trade-off in many real-world deployments.

8.2. Comparison with DP-FL

Table 5: summarises the trade-offs between our approach and DP-based secure FL.

Property	**Ours (Masking)**	**DP-FL**
Statistical cost (stats)	None	$\mathcal{O}(\Delta f/\varepsilon)$
Gradient privacy	Laplace DP (opt.)	Laplace/Gaussian DP
Estimation error	$\mathcal{O}(1/\sqrt{n})$	$\mathcal{O}(1/\sqrt{n} + b)$
Privacy model	HBC server	ε-DP
Client dropout	Requires recovery	Robust
Comm. overhead (setup)	$\mathcal{O}(n^2)$	$\mathcal{O}(n)$
Numerical precision	n/S	Exact

HBC = honest-but-curious; $b = \Delta f/\varepsilon$ = Laplace noise scale.

8.3. Empirical Validity of Theoretical Assumptions

The convergence guarantees of FedRényi rest on two assumptions: β-co-coercivity (Assumption 1) and L-Lipschitz prediction probability (Assumption 2), both validated empirically by (Ma et al., 2024) on the DUTCH dataset. Because Secure FedRényi passes only the aggregated statistics to the server (not any noise-corrupted version), both assumptions hold identically in our setting.

9. Limitations

Quadratic communication overhead. The pairwise DH key exchange requires $\mathcal{O}(n^2)$ messages in the setup phase, which is prohibitive for large-scale deployments ($n \sim 1{,}000$ clients). The spanning-tree construction of (Bell et al., 2020) reduces this to $\mathcal{O}(n \log n)$ and is a direct avenue for future work.

Client dropout. If a client drops out after transmitting its masked value, the corresponding mask term in all other clients' values becomes unmatched and the aggregation is incorrect. Robust dropout handling requires either a recovery protocol (surviving clients reveal their seeds with the dropped client) or threshold secret sharing (e.g., t-of-n Shamir shares), both of which add communication and computation overhead.

Honest-but-curious model. The privacy guarantee holds against a server that faithfully executes the protocol. Active adversaries (malicious servers or colluding clients) require stronger cryptographic tools such as zero-knowledge proofs or verifiable secret sharing, which we leave as future work.

Precision for large-scale systems. The quantisation error bound n/S grows with n. For $n = 1{,}000$ clients, $S = 10^6$ yields errors up to 10^{-3}, which may affect fairness estimation. Increasing S to 10^9 (requiring 64-bit integers) resolves this while remaining computationally feasible.

Scope of privacy. The masking protocol protects fairness *statistics* from the server. The Laplace DP layer protects gradient updates but introduces a fairness trade-off proportional to the noise scale $b = \Delta f/\varepsilon$. Membership inference and reconstruction attacks on the final model require complementary mechanisms.

10. Conclusion

We present Secure FedRényi, a privacy-preserving extension of the FedRényi federated fairness algorithm (Ma et al., 2024). Our framework incorporates two complementary approaches: (i) a Laplace differential privacy (DP) mechanism, and (ii) a secure aggregation method based on pairwise cryptographic masking derived from Diffie–Hellman key exchange (Diffie & Hellman, 1976).

Our empirical results show a clear distinction between these approaches. Applying Laplace DP introduces noise into the system, which perturbs the underlying gradient updates and leads to a degradation in fairness performance. In contrast, the proposed secure aggregation method preserves the original fairness exactly, as it protects the statistics without modifying their values. At the same time, it provides strong privacy guarantees against an honest-but-curious server.

These findings demonstrate that Secure FedRényi achieves both strong privacy and fairness preservation simultaneously, whereas DP-based methods incur a trade-off between privacy and fairness.

Three key results establish the value of the approach. **First**, Lemma 5.1 proves that the masking protocol recovers the exact global sum of statistics (modulo $\mathcal{O}(n/S)$ quantisation error), confirmed empirically with maximum errors of 1.91×10^{-6} and 2.26×10^{-6} in Rounds 1 and 2 respectively. **Second**, Corollary 5.3 and Proposition 5.4 establish that the $\mathcal{O}(1/\sqrt{n})$ estimation error and $\mathcal{O}(1/\varepsilon^4)$ convergence rate of FedRényi are preserved without modification for the masking layer—a zero-cost privacy guarantee. **Third**, our DP trade-off experiments show that at $\varepsilon \geq 5$ the Laplace DP noise has negligible impact on fairness, while at $\varepsilon \leq 0.1$ the fairness deviation remains within ± 0.003, quantifying the privacy–utility frontier.

Future work will scale the protocol to large n using spanning-tree mask cancellation (Bell et al., 2020), address client dropout via threshold secret sharing, and extend privacy guarantees to active adversarial settings.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically the intersection of privacy and algorithmic fairness in federated systems. The primary societal benefit is enabling fairness-aware AI systems in sensitive domains (healthcare, finance, social services) without requiring clients to reveal sensitive demographic information to the aggregating server. Potential risks include misuse of the protocol as a false assurance of privacy (it does not protect against active adversaries or provide full DP guarantees for model weights beyond the gradient noise layer), which we explicitly acknowledge in Section 9. We do not foresee other societal consequences that must be specifically highlighted here.

References

James Henry Bell, Keith Bonawitz, Adria Gascón, Tancrède Lepoint, and Mariana Raykova. Secure single-server aggregation with sublinear communication. *arXiv preprint arXiv:1909.05861*, 2020.

Keith Bonawitz, Vladimir Ivanov, Ben Kreuter, Antonio Marcedone, H. Brendan McMahan, Sarvar Patel, Daniel Ramage, Aaron Segal, and Karn Seth. Practical secure aggregation for privacy-preserving machine learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, pages 1175–1191. ACM, 2017.

Lingyang Chu, Lixin Wang, Yajie Dong, Jian Pei, Ziyun Zhou, and Yancheng Zhang. Fedfair: Training fair models in cross-silo federated learning. *arXiv preprint arXiv:2109.05658*, 2021.

Rachel Cummings, Varun Gupta, Dhamma Kimpara, and Jamie Morgenstern. On the compatibility of privacy and fairness. Pages 309–315, 2019.

Whitfield Diffie and Martin E. Hellman. New directions in cryptography. *IEEE Transactions on Information Theory*, 22(6):644–654, 1976.

Tian Li, Maziar Sanjabi, Ahmad Beirami, and Virginia Smith. Fair resource allocation in federated learning. In *International Conference on Learning Representations*, 2020.

Yifan Ma, Zhaowei Zhu, and Jian Pei. FedRényi: Federated learning with Rényi fairness. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, pages 1273–1282. PMLR, 2017.

Mehryar Mohri, Gary Sivek, and Ananda Theertha Suresh. Agnostic federated learning. In *Proceedings of the 36th International Conference on Machine Learning*, pages 4615–4625. PMLR, 2019.

Natalia Papadaki, Celestine Mendler-Dünner, and Martin Jaggi. Minimax demographic group fairness in federated learning. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pages 142–159. ACM, 2022.